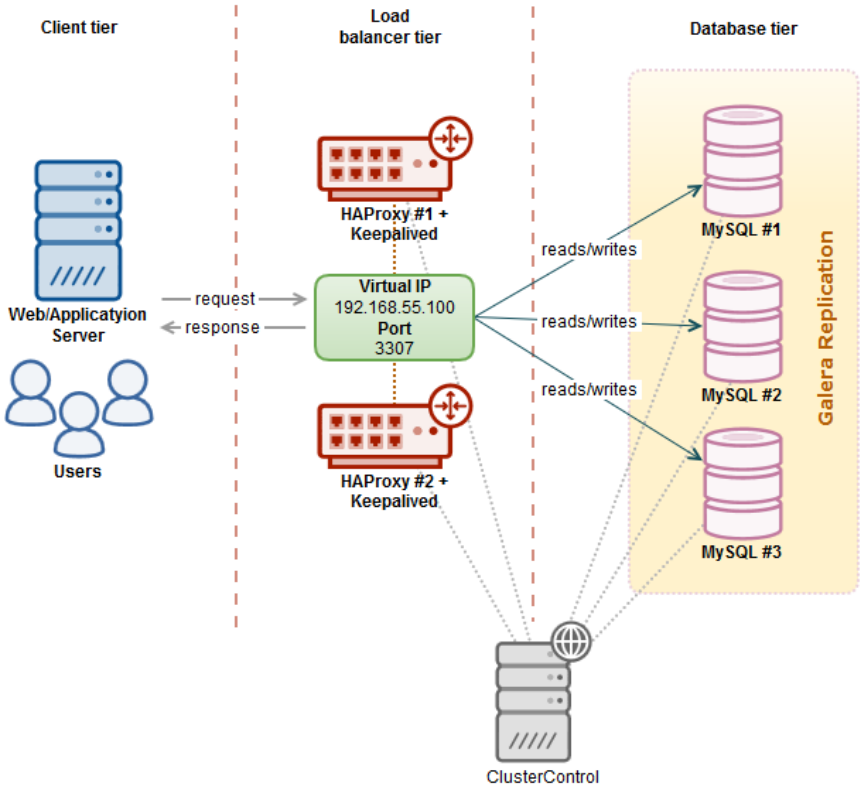
Implementasi High Availability (HA) MySQL InnoDB Cluster dengan Orkestrasi HAProxy dan Keepalived

Topologi sistem

Sistem ini mengadopsi arsitektur *three-tier* yang dirancang untuk mengeliminasi *Single Point of Failure* (SPOF). Komponen utama penyusun topologi ini meliputi:

- **Tier Klien**: Terdiri dari pengguna dan *Application Server* yang melakukan transmisi permintaan (*request*) dan menerima respons (*response*).
- **Tier Load Balancer**: Mengimplementasikan redundansi perangkat lunak menggunakan dua unit HAProxy yang disinergikan dengan Keepalived. Komponen ini menyediakan mekanisme *failover* melalui penggunaan *Virtual IP* (VIP) 192.168.55.100 pada port 3307.
- **Tier Basis Data**: Menggunakan tiga node MySQL yang terintegrasi dalam mekanisme *Galera Replication* atau *InnoDB Cluster* untuk menjamin konsistensi data antar node.



1. Verifikasi Integrasi Virtual IP (VIP)
   Berdasarkan inspeksi pada antarmuka jaringan ens33 di node mysql1, sistem telah berhasil mengalokasikan beberapa alamat IP virtual:

- **Alamat IP Statis**: 10.20.10.11/24.

- **Alamat IP Virtual**: Terdeteksi adanya VIP 10.20.10.100, 10.20.10.101, dan 10.20.10.102 dengan subnet mask /32.

```
root@mysql1:~# ip addr show ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d7:71:5f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 10.20.10.11/24 brd 10.20.10.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet 10.20.10.100/32 scope global ens33
       valid_lft forever preferred_lft forever
    inet 10.20.10.101/32 scope global ens33
       valid_lft forever preferred_lft forever
    inet 10.20.10.102/32 scope global ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fed7:715f/64 scope link
       valid_lft forever preferred_lft forever
root@mysql1:~#
```

2. Remote Privileges
   Otentikasi pada *database engine* telah dikonfigurasi untuk mendukung monitoring eksternal. Terdapat akun haproxy_check dengan host % (wildcard) yang memungkinkan HAProxy melakukan inspeksi kesehatan (*health check*) terhadap ketersediaan layanan MySQL

```
root@mysql1:~# mysql -u root -p -e "SELECT User, Host FROM mysql.user WHERE User IN ('root', 'haproxy_check');"
Enter password:
+---------------+-----------+
| User          | Host      |
+---------------+-----------+
| haproxy_check | %         |
| root          | %         |
| root          | localhost |
+---------------+-----------+
root@mysql1:~#
```

3. Inspeksi Konsensus Quorum pada InnoDB Cluster
   Evaluasi terhadap integritas klaster dilakukan melalui fungsi cluster.status(), dengan rincian operasional sebagai berikut:

- **Status Klaster**: Tercatat **ONLINE** dengan toleransi kegagalan maksimum satu node (*ONE failure*).
- **Modus Operasional**: *Single-Primary*.
- **Topologi Node**:
  - **Node mysql1**: Berperan sebagai **PRIMARY** dengan modus operasional *Read/Write* (R/W).
  - **Node mysql2 & mysql3**: Berperan sebagai **SECONDARY** dengan modus operasional *Read-Only* (R/O).
  - **Versi Perangkat Lunak**: Seluruh node menjalankan MySQL versi 8.0.45.

```
MySQL 10.20.10.11:3306 ssl Py > cluster.status()
{
    "clusterName": "my_cluster",
    "defaultReplicaSet": {
        "name": "default",
        "primary": "mysql1:3306",
        "ssl": "REQUIRED",
        "status": "OK",
        "statusText": "Cluster is ONLINE and can tolerate up to ONE failure.",
        "topology": {
            "mysql1:3306": {
                "address": "mysql1:3306",
                "memberRole": "PRIMARY",
                "mode": "R/W",
                "readReplicas": {},
                "replicationLag": "applier_queue_applied",
                "role": "HA",
                "status": "ONLINE",
                "version": "8.0.45"
            },
            "mysql2:3306": {
                "address": "mysql2:3306",
                "memberRole": "SECONDARY",
                "mode": "R/O",
                "readReplicas": {},
                "replicationLag": "applier_queue_applied",
                "role": "HA",
                "status": "ONLINE",
                "version": "8.0.45"
            },
            "mysql3:3306": {
                "address": "mysql3:3306",
                "memberRole": "SECONDARY",
                "mode": "R/O",
                "readReplicas": {},
                "replicationLag": "applier_queue_applied",
                "role": "HA",
                "status": "ONLINE",
                "version": "8.0.45"
            }
        },
        "topologyMode": "Single-Primary"
    },
    "groupInformationSourceMember": "mysql1:3306"
}
MySQL 10.20.10.11:3306 ssl Py >
```

4. Status Operasional *Layer 4 Load Balancer*

Layanan HAProxy pada layer 4 diidentifikasi berada dalam status **active (running)**. Pengujian perutean (*routing test*) dari node mysql3 ke VIP 10.20.10.100 pada port 3307 mengonfirmasi bahwa koneksi berhasil diteruskan secara transparan ke node master (mysql1) pada port standar 3306.

```
root@mysql1: ~
root@mysql1:~# sudo systemctl status haproxy --no-pager
● haproxy.service - HAProxy Load Balancer
     Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: ena
bled)
     Active: active (running) since Tue 2026-03-31 03:22:45 UTC; 1h 20min ago
       Docs: man:haproxy(1)
             file:/usr/share/doc/haproxy/configuration.txt.gz
   Main PID: 1328 (haproxy)
     Status: "Ready."
      Tasks: 3 (limit: 4544)
     Memory: 43.5M (peak: 44.0M)
        CPU: 6.571s
     CGroup: /system.slice/haproxy.service
             ├─1328 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/hapr
oxy.pid -S /run/h…
             └─1361 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/hapr
oxy.pid -S /run/h…

Mar 31 04:42:48 mysql1 haproxy[1361]: 10.20.10.2:21344 [31/Mar/2026:04:42:48.517]
stats stats…P/1.1"
Mar 31 04:42:53 mysql1 haproxy[1361]: 10.20.10.2:21344 [31/Mar/2026:04:42:53.552]
stats stats…P/1.1"
Mar 31 04:42:58 mysql1 haproxy[1361]: 10.20.10.2:21344 [31/Mar/2026:04:42:58.593]
stats stats…P/1.1"
Mar 31 04:43:03 mysql1 haproxy[1361]: 10.20.10.2:21344 [31/Mar/2026:04:43:03.640]
stats stats…P/1.1"
Mar 31 04:43:08 mysql1 haproxy[1361]: 10.20.10.2:21344 [31/Mar/2026:04:43:08.679]
stats stats…P/1.1"
Mar 31 04:43:13 mysql1 haproxy[1361]: 10.20.10.2:21344 [31/Mar/2026:04:43:13.715]
stats stats…P/1.1"
Mar 31 04:43:18 mysql1 haproxy[1361]: 10.20.10.2:21344 [31/Mar/2026:04:43:18.762]
stats stats…P/1.1"
Mar 31 04:43:23 mysql1 haproxy[1361]: 10.20.10.2:21344 [31/Mar/2026:04:43:23.853]
stats stats…P/1.1"
Mar 31 04:43:28 mysql1 haproxy[1361]: 10.20.10.2:21344 [31/Mar/2026:04:43:28.902]
stats stats…P/1.1"
Mar 31 04:43:33 mysql1 haproxy[1361]: 10.20.10.2:21344 [31/Mar/2026:04:43:33.944]
stats stats…P/1.1"
Hint: Some lines were ellipsized, use -l to show in full.
root@mysql1:~#
```

```
root@mysql2: ~
root@mysql2:~# sudo systemctl status haproxy --no-pager
● haproxy.service - HAProxy Load Balancer
     Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-03-31 03:22:40 UTC; 1h 21min ago
       Docs: man:haproxy(1)
             file:/usr/share/doc/haproxy/configuration.txt.gz
   Main PID: 1023 (haproxy)
     Status: "Ready."
      Tasks: 3 (limit: 4544)
     Memory: 44.7M (peak: 45.2M)
        CPU: 4.739s
     CGroup: /system.slice/haproxy.service
             ├─1023 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/h…
             └─1275 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/h…

Mar 31 03:22:43 mysql2 haproxy[1275]: proxy mysql-cluster has no server available!
Mar 31 03:22:47 mysql2 haproxy[1275]: [WARNING]  (1275) : Backup Server mysql-cluster/mysql2 …queue.
Mar 31 03:22:47 mysql2 haproxy[1275]: Backup Server mysql-cluster/mysql2 is UP, reason: Layer…queue.
Mar 31 03:22:47 mysql2 haproxy[1275]: Backup Server mysql-cluster/mysql2 is UP, reason: Layer…queue.
Mar 31 03:22:50 mysql2 haproxy[1275]: [WARNING]  (1275) : Server mysql-cluster/mysql1 is UP, …queue.
Mar 31 03:22:50 mysql2 haproxy[1275]: Server mysql-cluster/mysql1 is UP, reason: Layer4 check…queue.
```

```
root@mysql3: ~
root@mysql3:~# sudo systemctl status haproxy --no-pager
● haproxy.service - HAProxy Load Balancer
     Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-03-31 03:22:50 UTC; 1h 21min ago
       Docs: man:haproxy(1)
             file:/usr/share/doc/haproxy/configuration.txt.gz
   Main PID: 1098 (haproxy)
     Status: "Ready."
      Tasks: 3 (limit: 4544)
     Memory: 44.7M (peak: 45.2M)
        CPU: 5.081s
     CGroup: /system.slice/haproxy.service
             ├─1098 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/h…
             └─1221 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/h…

Mar 31 03:22:50 mysql3 haproxy[1098]: [WARNING]  (1098) : config : parsing [/etc/haproxy/hapr…plog'.
Mar 31 03:22:50 mysql3 haproxy[1098]: [NOTICE]   (1098) : New worker (1221) forked
Mar 31 03:22:50 mysql3 haproxy[1098]: [NOTICE]   (1098) : Loading success.
Mar 31 03:22:50 mysql3 systemd[1]: Started haproxy.service - HAProxy Load Balancer.
Mar 31 03:22:51 mysql3 haproxy[1221]: [WARNING]  (1221) : Backup Server mysql-cluster/mysql3 …queue.
Mar 31 03:22:51 mysql3 haproxy[1221]: Backup Server mysql-cluster/mysql3 is DOWN, reason: Lay…queue.
```

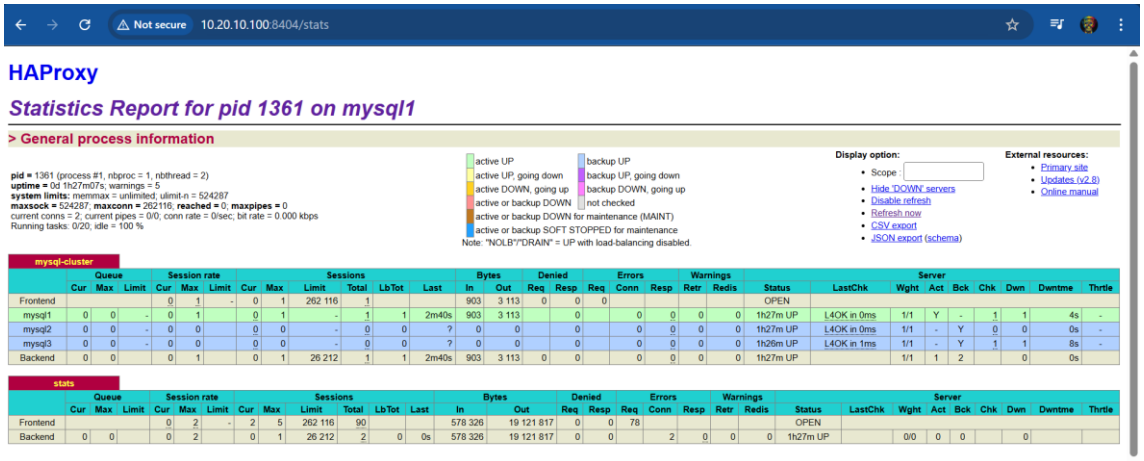
5. Routing Test

```
root@mysql3:~# mysql -u root -p -h 10.20.10.100 -P 3307 -e "SELECT @@hostname AS 'Active_Master_Node
', @@port AS 'Target_Port';"
Enter password:
+--------------------+-------------+
| Active_Master_Node | Target_Port |
+--------------------+-------------+
| mysql1             |        3306 |
+--------------------+-------------+
root@mysql3:~#
```

6. Matriks Telemetri Visual

Eksperimen dilakukan dengan menghentikan layanan Keepalived pada node mysql1 secara paksa (sudo systemctl stop keepalived). Hasil observasi menunjukkan:

- Terjadi transisi kepemilikan alamat IP virtual secara otomatis.
- Node mysql2 berhasil mengambil alih fungsi VIP (10.20.10.100, 10.20.10.101, 10.20.10.102) pada antarmuka jaringan ens33.
- Hal ini membuktikan efektivitas protokol VRRP (*Virtual Router Redundancy Protocol*) dalam menjaga ketersediaan layanan saat terjadi anomali pada node utama.



7. Simulasi *Failover* dan Transisi Status

```
root@mysql1:~# sudo systemctl stop keepalived
root@mysql1:~#
```

```
root@mysql2:~# ip addr show ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1d:b8:ab brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 10.20.10.12/24 brd 10.20.10.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet 10.20.10.100/32 scope global ens33
       valid_lft forever preferred_lft forever
    inet 10.20.10.101/32 scope global ens33
       valid_lft forever preferred_lft forever
    inet 10.20.10.102/32 scope global ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe1d:b8ab/64 scope link
       valid_lft forever preferred_lft forever
root@mysql2:~# |
```